

RIVIAN

Sanjith Menon Paint ME Co-op



R1 Hood Jig Redesign



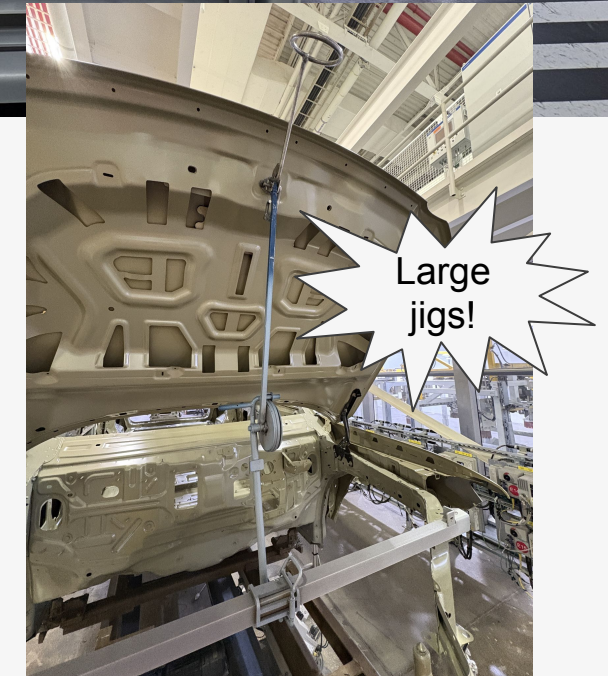
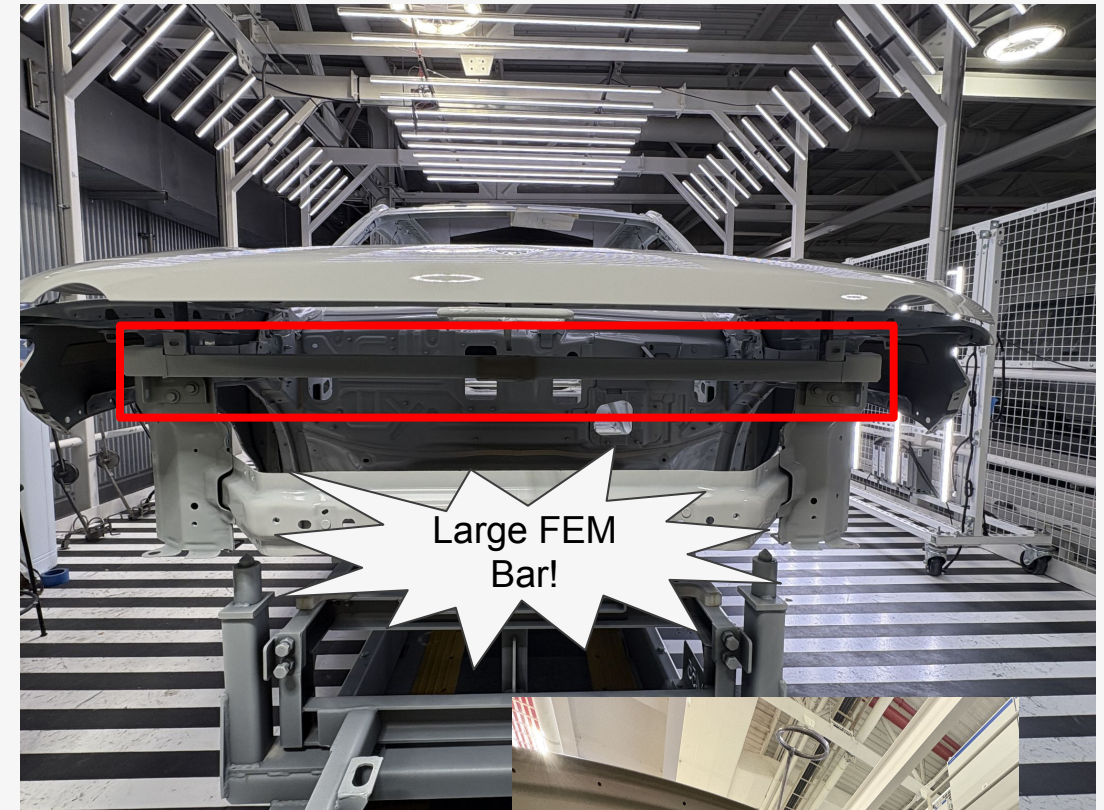
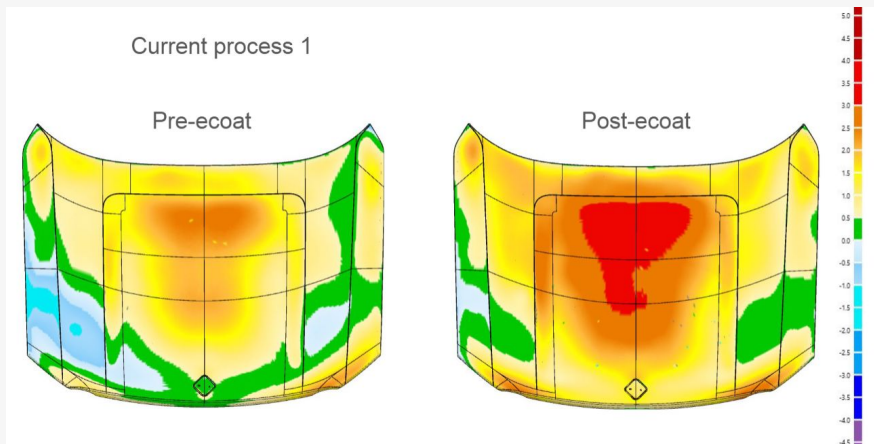
Problem

Initial RC

- FEM Brackets are being damaged by large FEM bar used for e-coat jigs

Goals

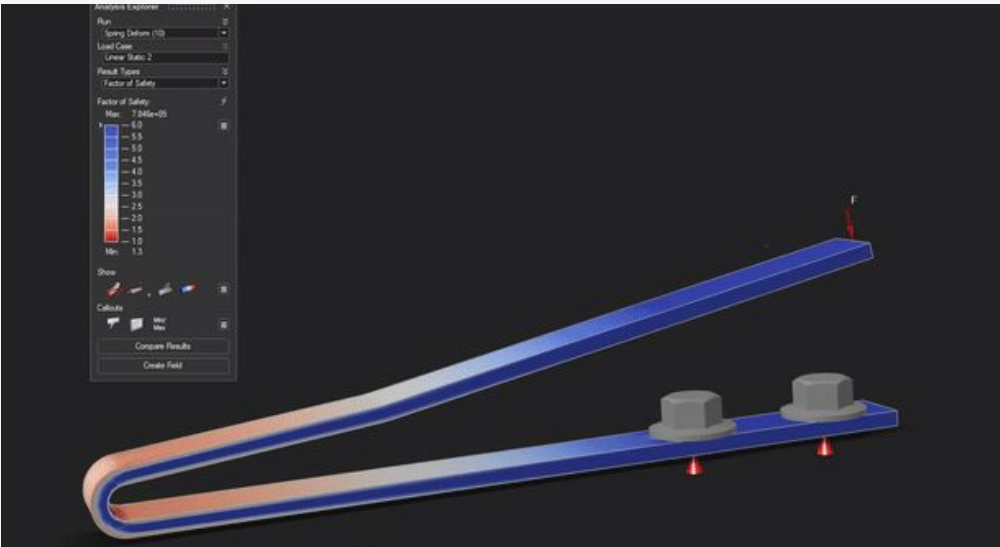
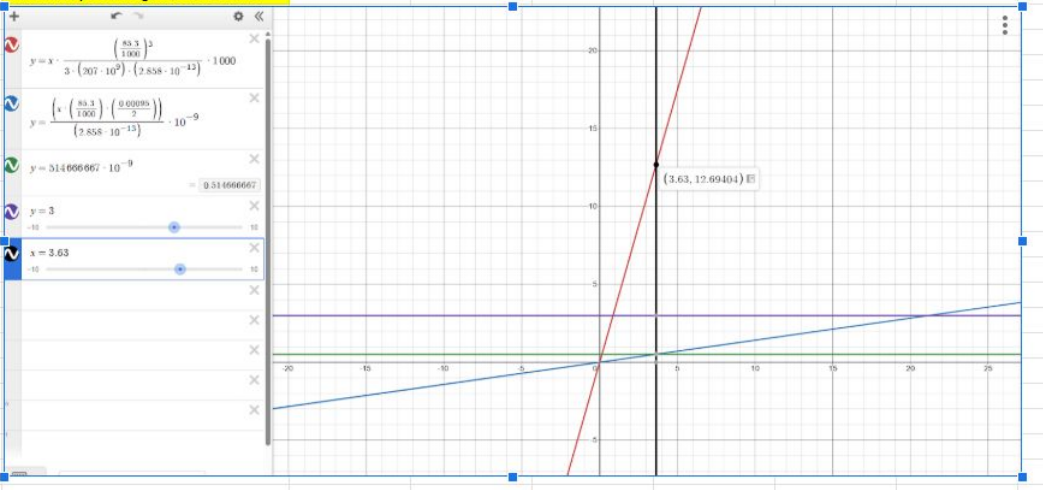
- Eliminate deformation in FEM brackets
- Maintain same hood angle through e-coat and paint to reduce jig change processes
- Improve e-coat quality by identifying a hood angle that reduces bridging defects
- Reduce hood deformation seen after PTED process



Design Process

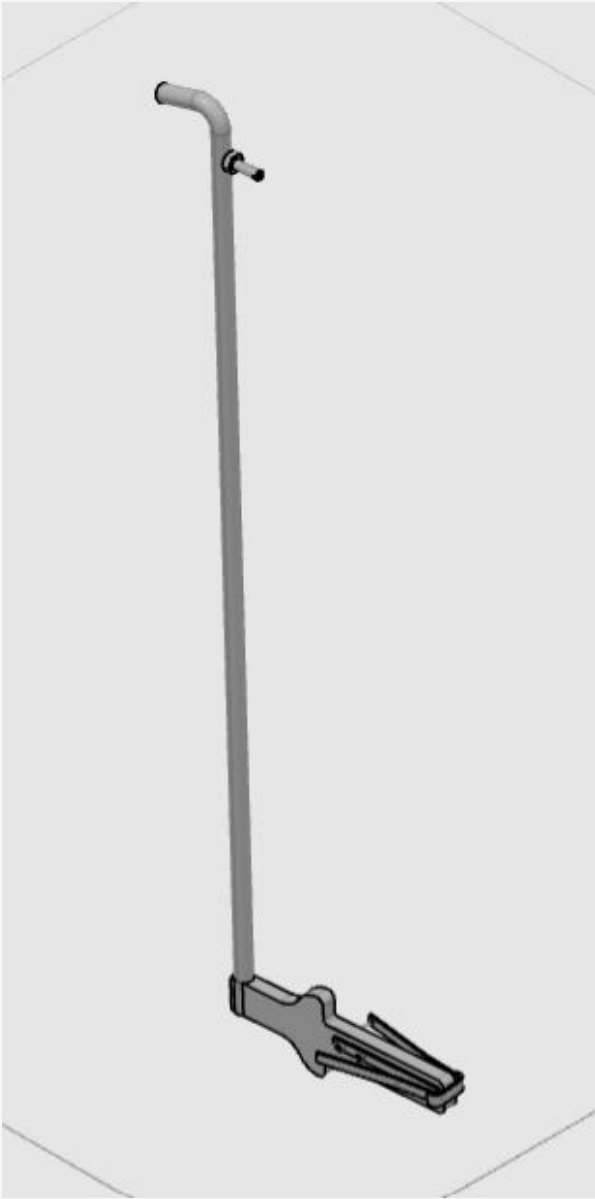


deflection plotted against force in N



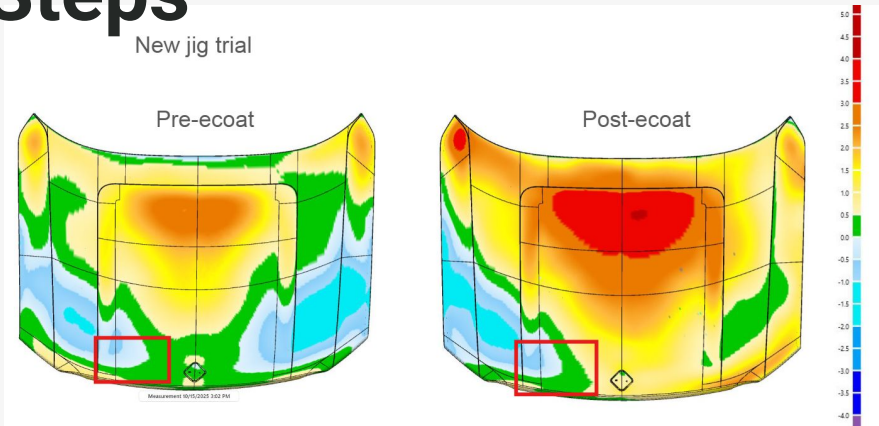
Final Solution and Results

Overall Paint/Body/GA Process - R1 Hood Jigging				
Type	New Condition	Old Condition	Net	Percent Change
Total fasteners used	4	6	-2	-33%
Distinct Jig Pieces	3	4	-1	-25%
Large Pieces	0	2	-2	-100%
Small or Medium Pieces	4	3	+1	+25%



 Reduced hood deformation - results of trial coming soon

Next Steps



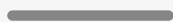
Order jigs from supplier

SOW is ready and CAD package has been compiled.



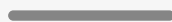
Confirm results of hood deformation trial

ATOS scan results will show deformation reduction by 12/3



New hood angle for paint quality

Investigating a new hood angle for paint quality, trialed through e-coat to ensure quality there, need further trials in medium run to confirm



Adjust jig height

Once new angle is identified, validated, and bought off; jig height is a simple change

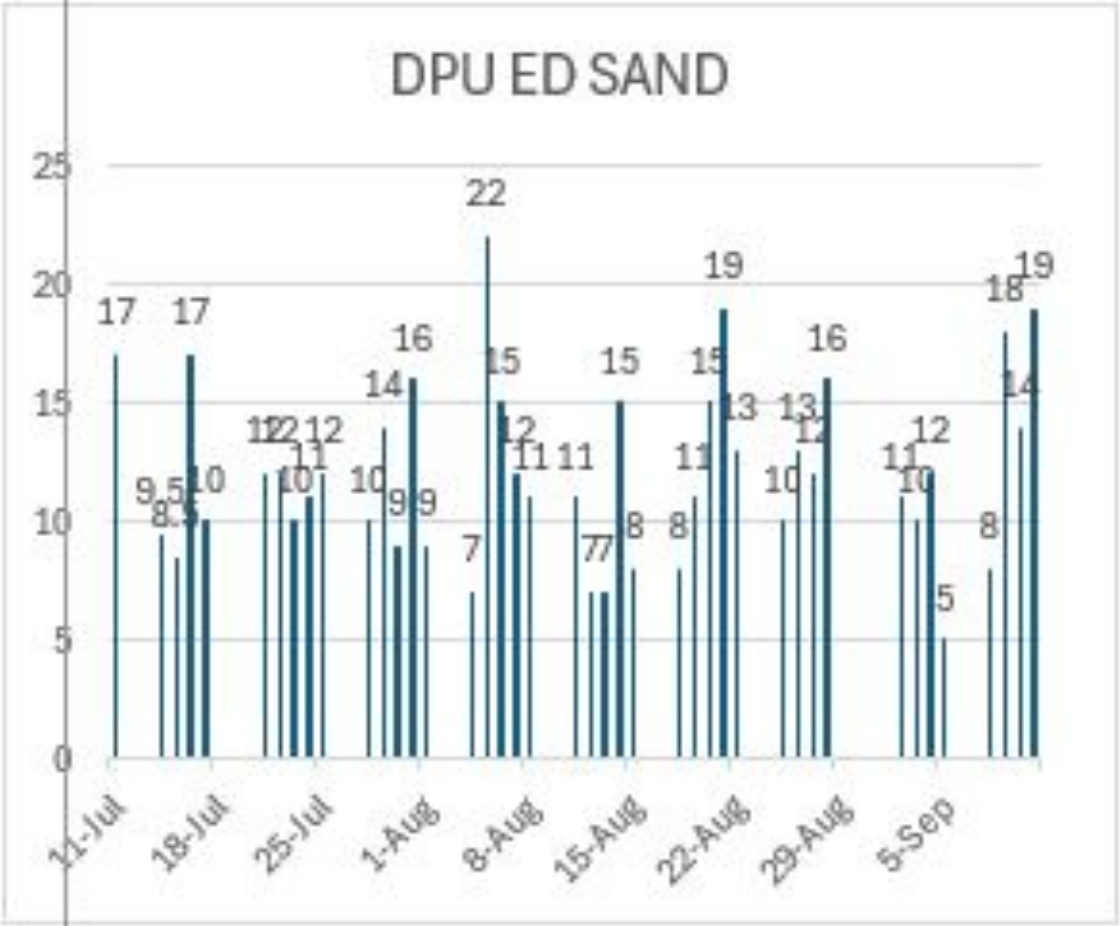


ED Blow Off

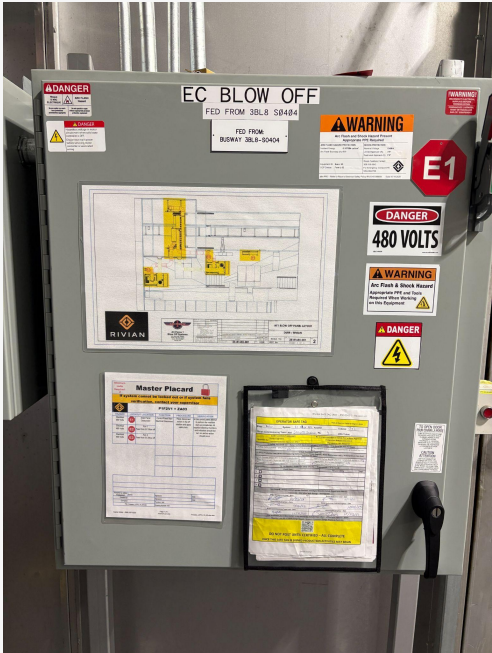
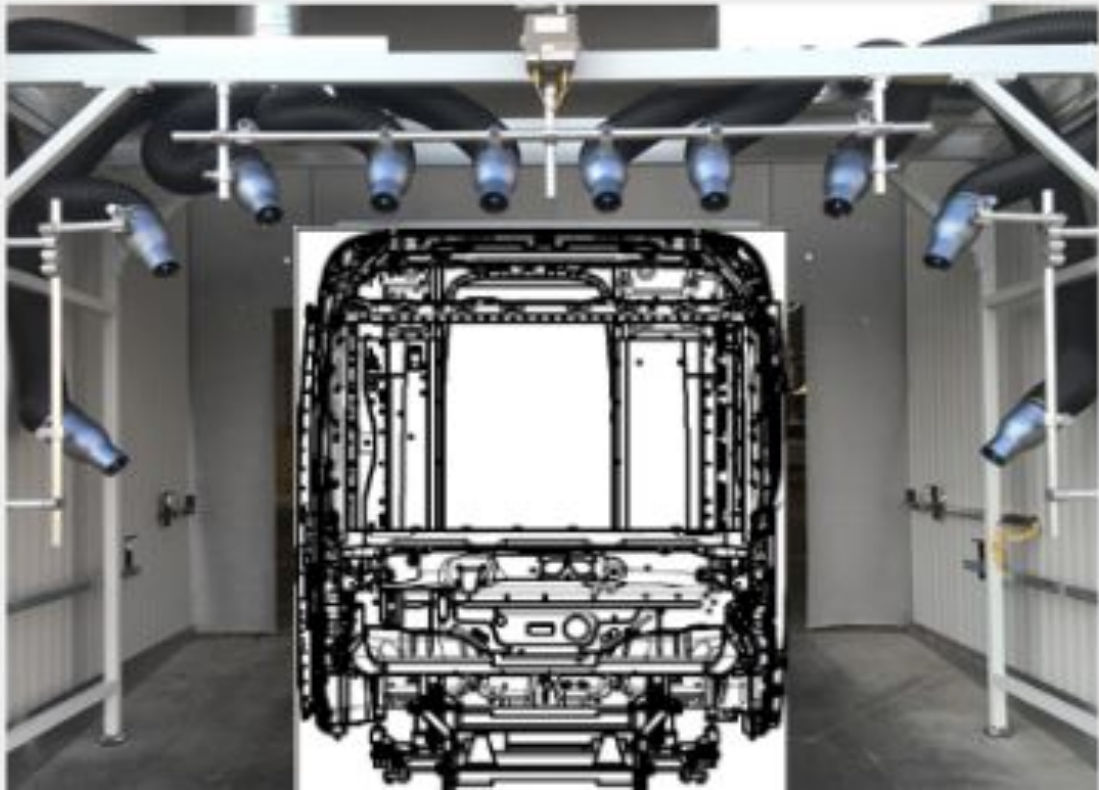


Problem/Context

- Drip defects at ED sand causing increase in cycle time and DPU
- Drips are caused by excess e-coat build-up in cavities of the vehicles which boil out in the ovens and drip on surfaces



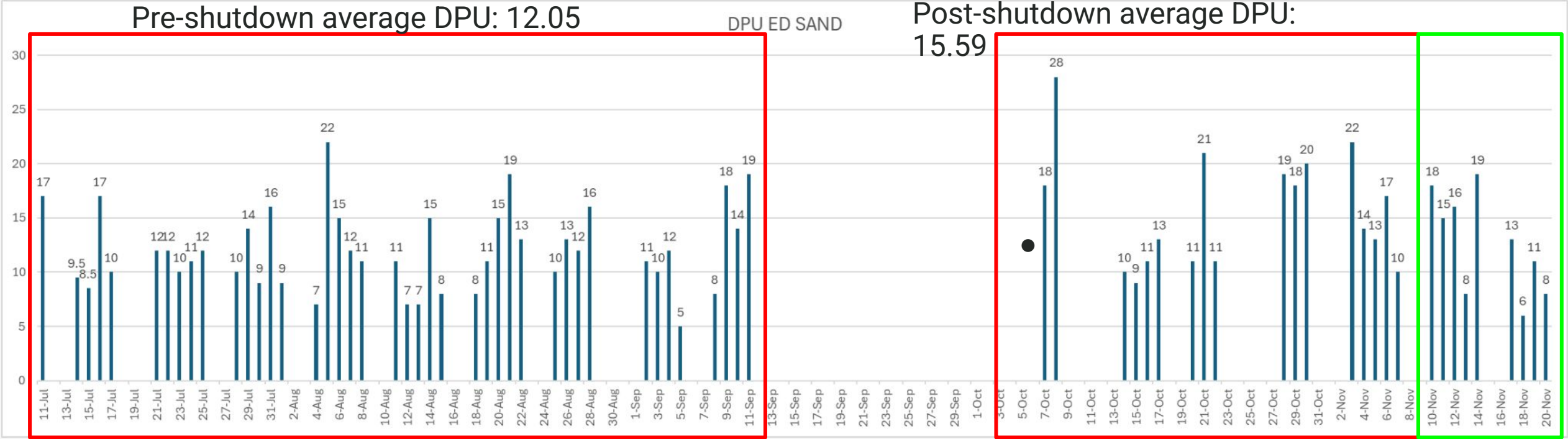
Solution, Install, Commissioning, Validation



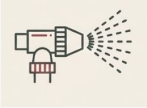
Results

- Not yet fully pointed the nozzles at problem areas
- This should decrease DPU significantly
- Also DPU includes many defects not just boil out which is what the blowers target so need to confirm if boil outs are still the top defect or not

Post-blower
average
DPU: 12.67



Next Steps



Position nozzles to target problem areas



Analyze results of EHS test



Sign Green Tag!



Feather Duster



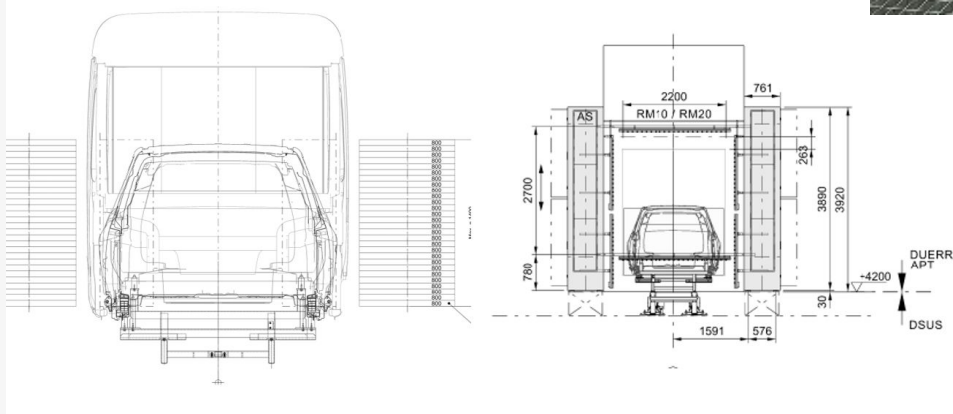
Problem

- Flat rollers don't contact parts of the contoured R1 body
- Home position is set to EDV height - but EDV doesn't even go through this station

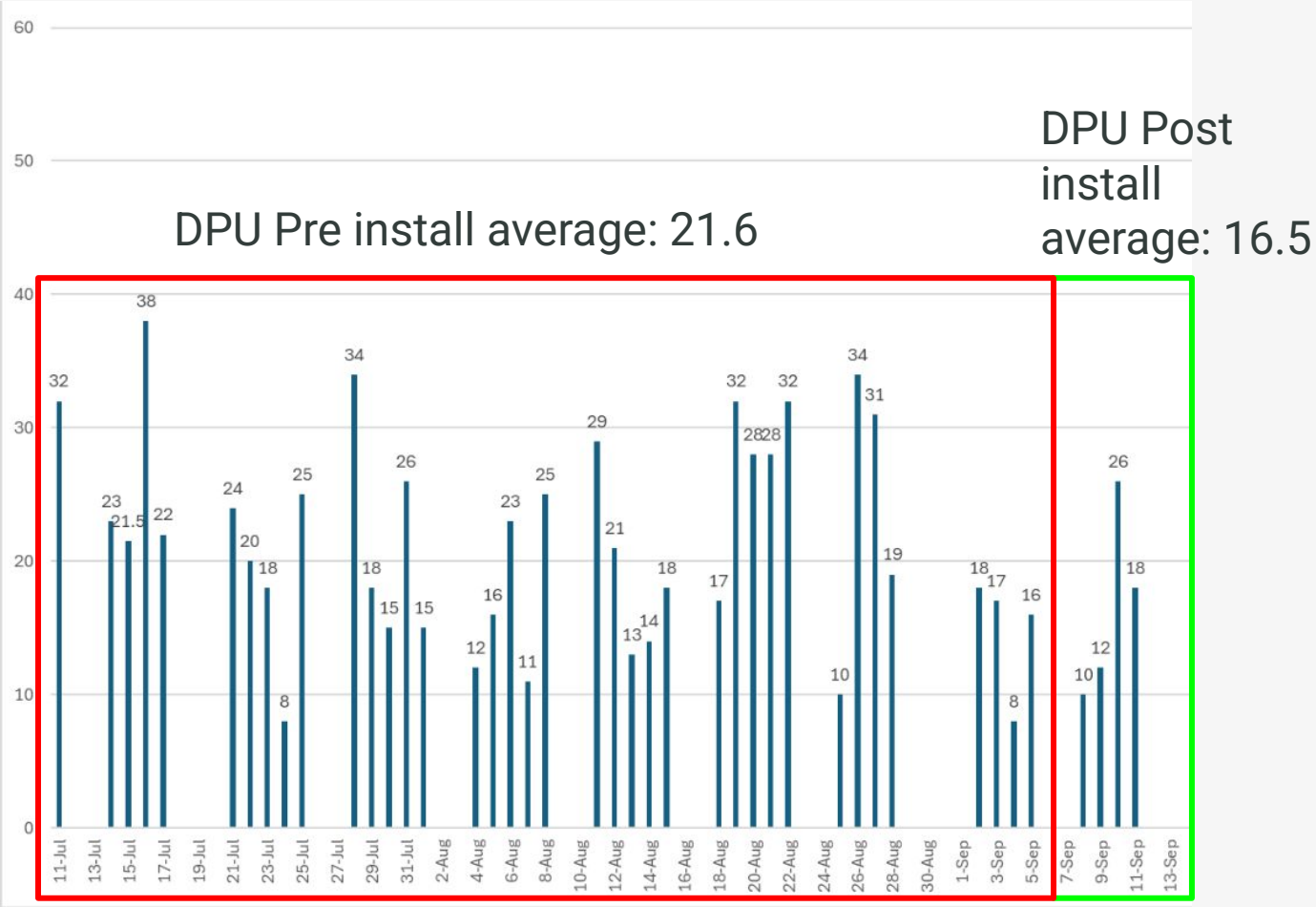


Feather Duster

- Ready position lowered
- Rollers contoured
- Crush amount specified



Results



Skid Extension (Flipper) Station



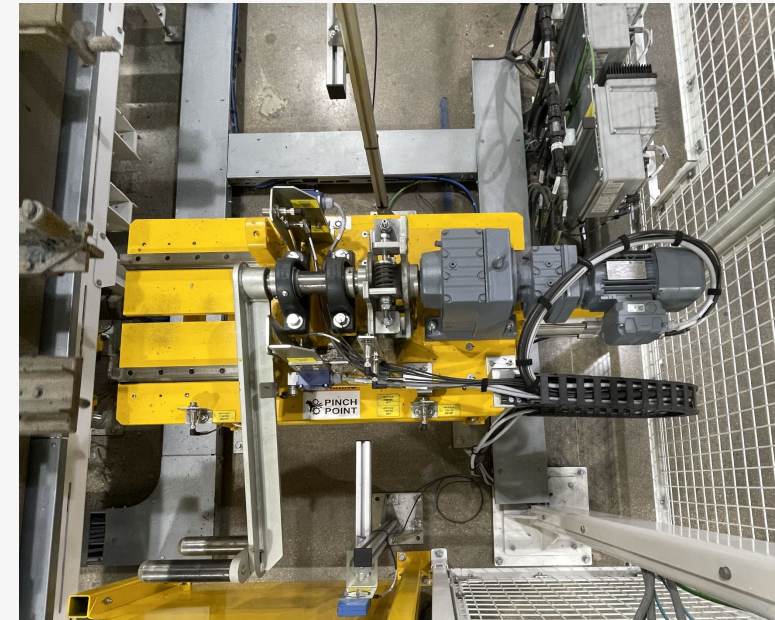
Problem

Context

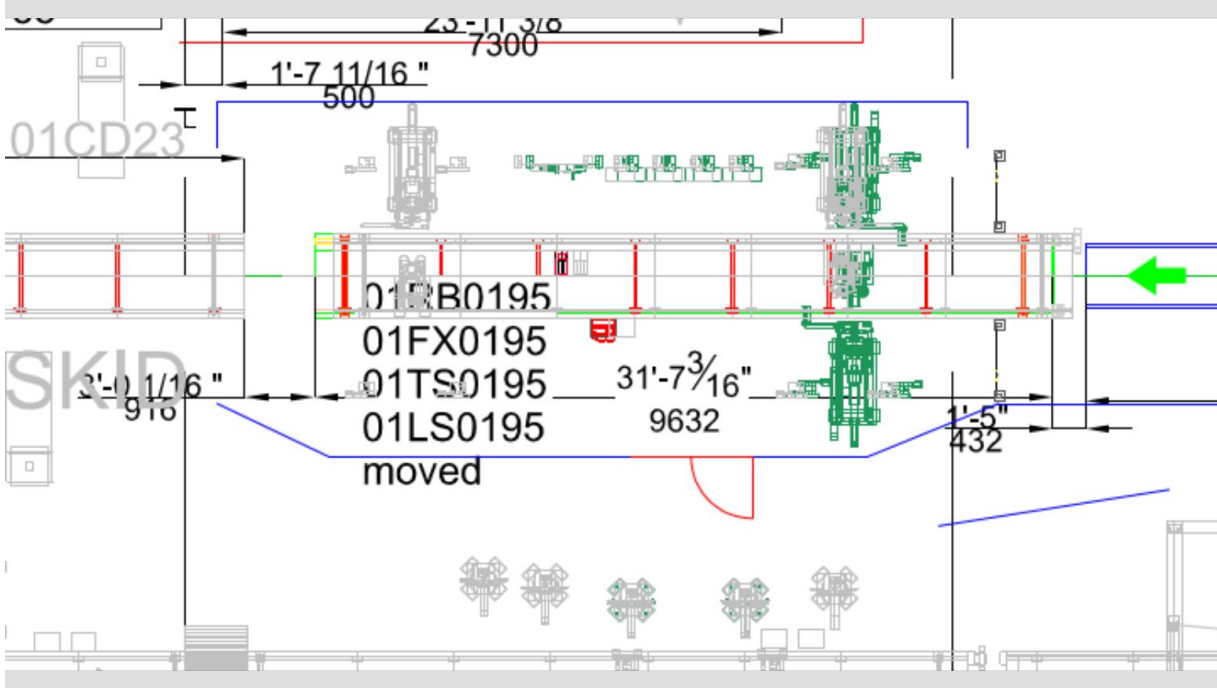
- Skid flipper required to change configuration of e-coat skids between EDV and R1 models

Failures

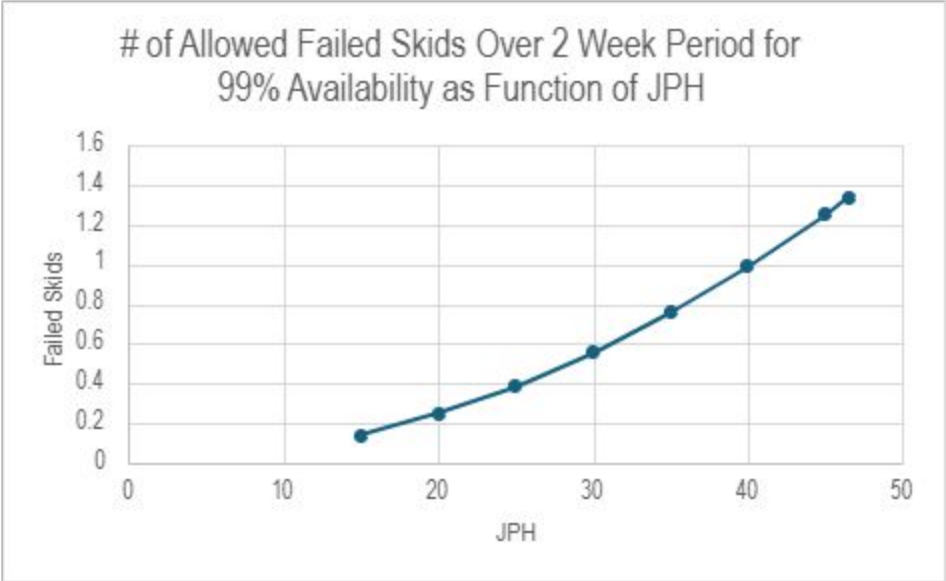
- E-coat gets stuck in pivot points of extension and causes flipper to fault because the extension is stuck
- Manual intervention required



Solution

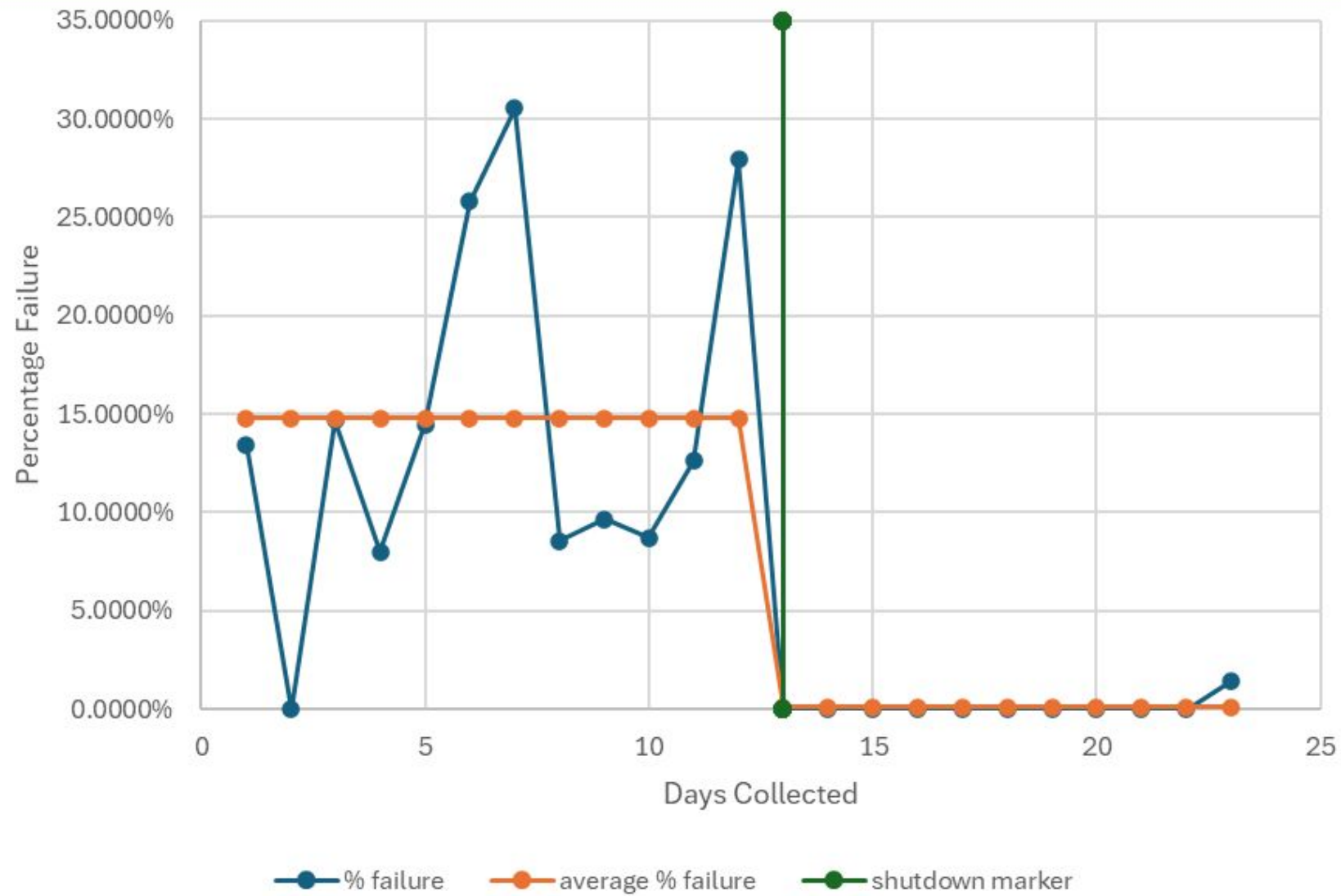


Validating success

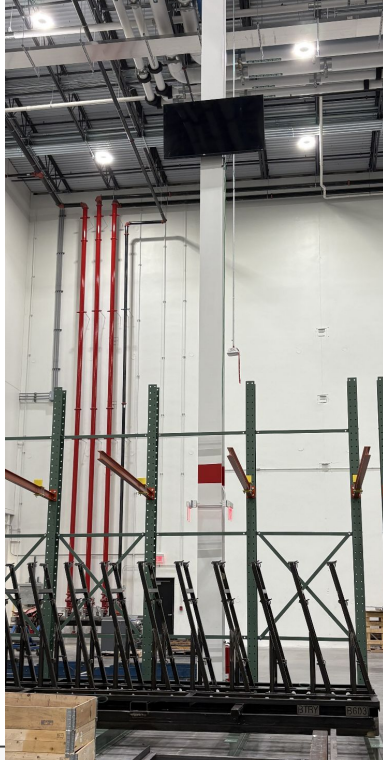


Results

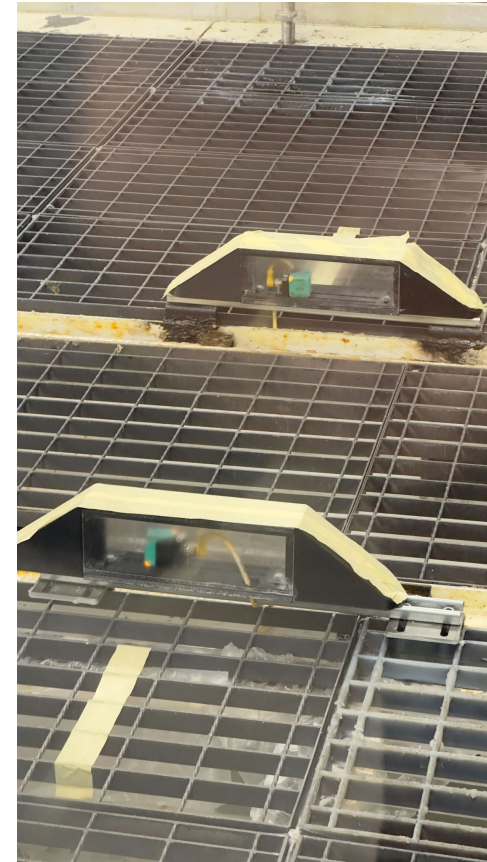
Pre shutdown	range from 8/29 - 9/12	
# of faults	# of required flips	percentage failure
126	851	0.148061105
		14.8%
Post shutdown	range from 11/14 - 11/25	
# of faults		
2	1742	0.001148106
		0.115%



Other Projects



Andon Screen Install for Parts PTED -
one temporarily up, facilities and IT working
on permanents



Prox Switch Cover redesign - installed 4,
3 more switches required modifications to
cover



Other Projects

Destacker modifications for new paint skids - found root cause, had Durr implement changes quick

Skateboard skid modifications - coordinated adding fork tubes to R1 skateboard skids

Security Camera Audit - identified which cameras exist, which need to exist, create folder structures

Confined Spaces Identification - walked the shop and identified new confined spaces without placard

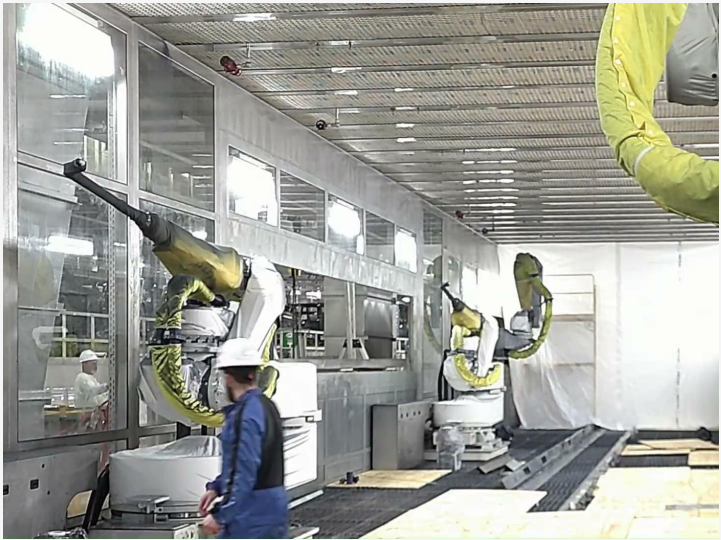
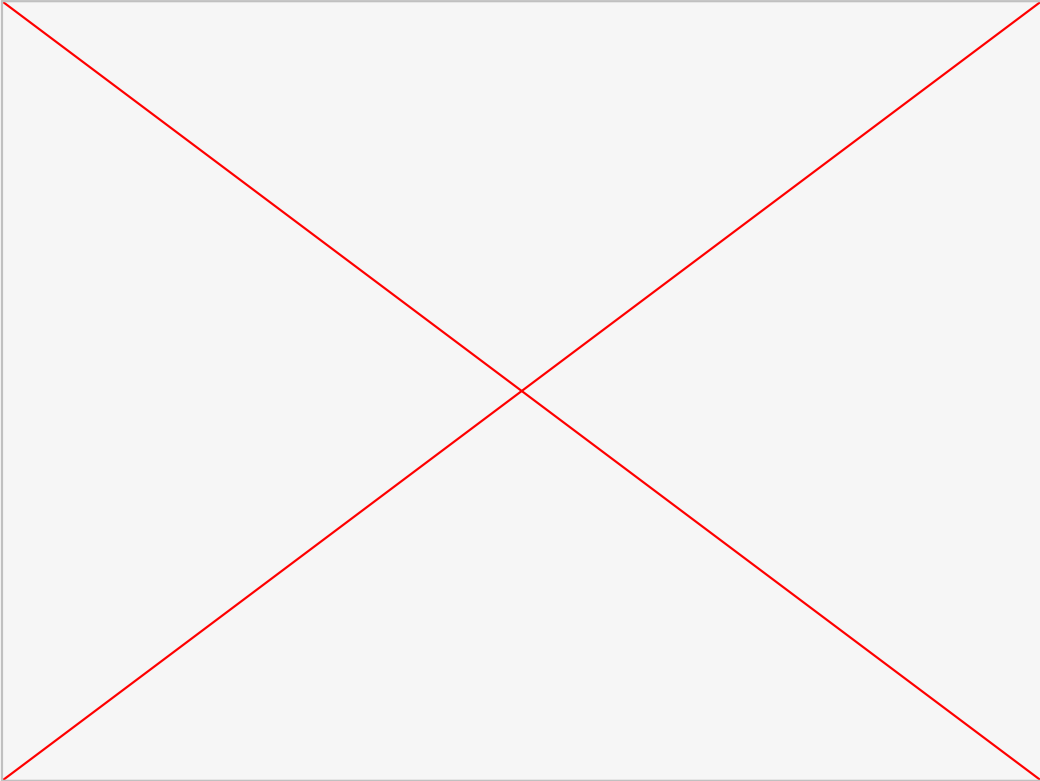
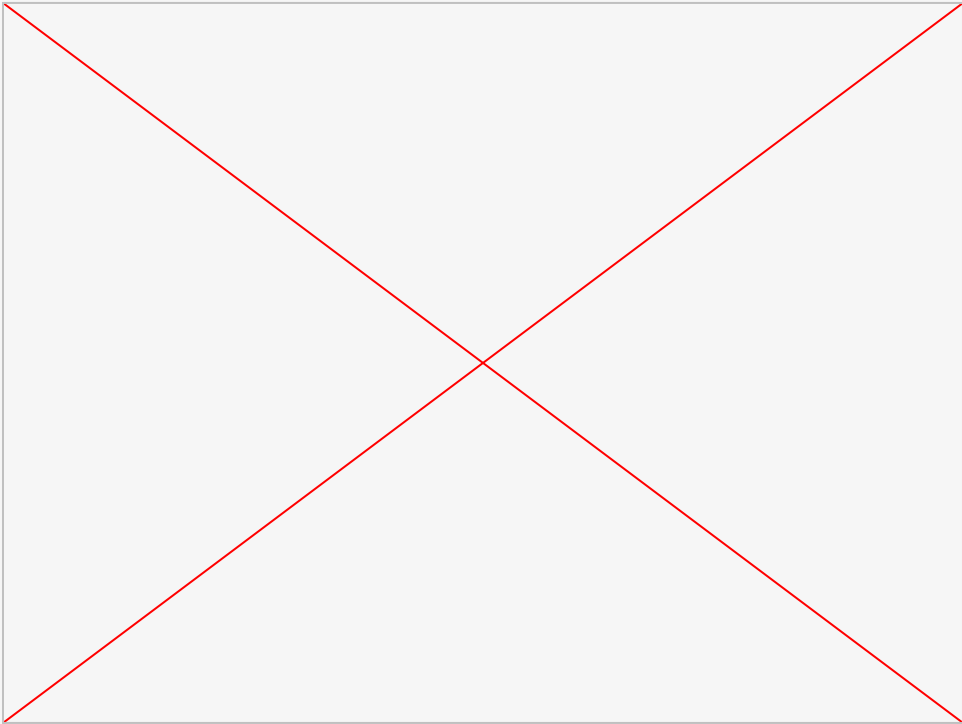
Grating replacement in paint booths - measured sizes, requested quotes from suppliers, and ordered new grates for paint booths for safety

Safety portal modifications for new paint skids - identified problem areas and had Durr implement changes

Weekend Work - jigging, validation, etc.



Timelapse Videos



Minor Improvements



Alignment with all stakeholders first - even with rushed deadlines. EX. Parts PTED Andon

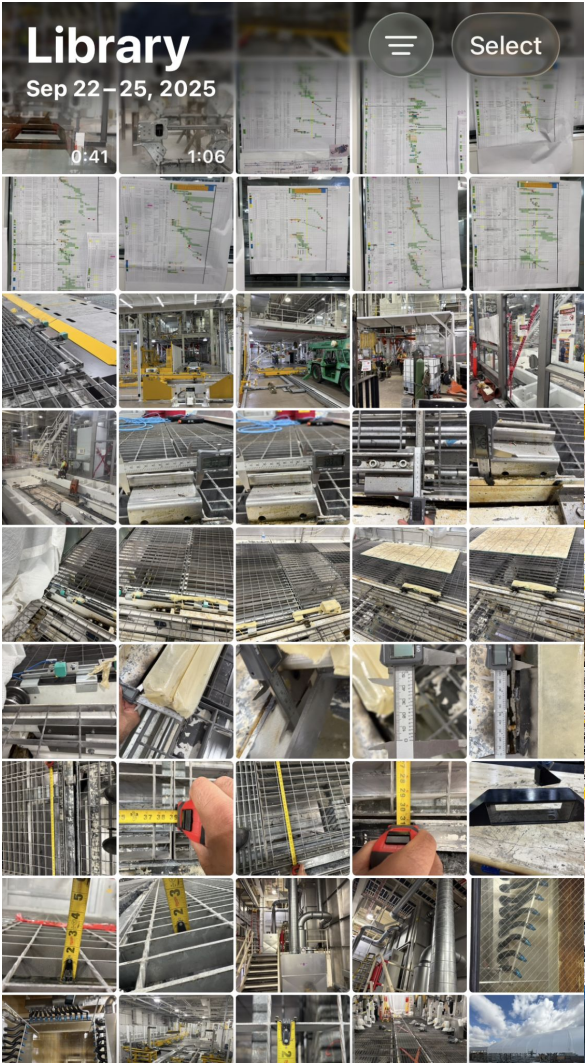


Get quality buyoff and support from appropriate stakeholders ahead of time. EX. ED Blow Off



Overall Experience

NORMAL, ILLINOIS



Overall Experience

TRAVEL

